

Mannan Shukla

Sunnyvale, California • US Citizen

(609) 575-0436 • mannanshukla@icloud.com • linkedin.com/in/mannanshukla • github.com/mannanshukla

TECHNICAL SKILLS

Languages: Python, Go, Bash, TypeScript

Inference: vLLM, SGLang, TensorRT, NVIDIA Dynamo, CUDA, NCCL, PagedAttention, KV cache, continuous batching

ML & Agents: PyTorch, Hugging Face, LoRA, LangGraph, MCP, KubeFlow

Infrastructure: Kubernetes, Docker, Helm, Terraform, AWS, ArgoCD, Cilium, Slurm

Production: Postgres, Prometheus, Grafana, GitHub Actions, CI/CD, GitOps

EXPERIENCE

Software Engineer

Voltage Park / Lightning AI • San Francisco, CA & Redmond, WA

July 2025 - Present

- Shipped hot model-weight reloading for **vLLM** inference deployments via a **Go** sidecar, cutting fine-tuned model rollout time from 10-30 minutes to seconds without restarting GPU workers
- Designed an async/batch inference platform with **Postgres**-backed task durability, pull-based GPU workers, and queue-depth autoscaling, enabling burst handling and scale-to-zero for idle workloads
- Built core pieces of an **S3**-compatible object store (**VAST**-backed): a health-aware **P2C** client-side load balancer over CNode IPs with lock-free in-flight tracking and automatic failover, plus **Terraform/Vault**-managed **RDS** infra
- Built distributed inference orchestration in **Go** on **Kubernetes**, implementing deployment-target controllers and **DynamoDB** event sourcing to reconcile and recover fleets of GPU-serving replicas
- Created the **vp-cloud-sdk Python** SDK for Voltage Park's orchestration platform, exposing fleet/node/remediation APIs and a full staging release pipeline with **GitHub Actions**
- Own a multi-tenant GPU platform spanning **ArgoCD**, **Terraform**, **Helm**, **Cilium/Envoy**, **NVIDIA GPU Operator**, and **Prometheus/Grafana**, with tenant isolation and production observability
- Built a **Slurm** GPU health-check suite for distributed training (**ECC**, **PCIe AER**, **NCCL** stalls, **DCGM** checks, thermal) that auto-drains unhealthy nodes, creates RMA tickets, and sends **Slack** alerts to protect multi-node jobs
- Operate production **H100** clusters and resolve incidents across GPU scheduling, distributed networking, model storage, and **NCCL** workloads spanning Kubernetes and infrastructure layers

Undergraduate Researcher: AIOS

Rutgers University • New Brunswick, NJ

Apr 2024 - Apr 2025

- Built an **LLM** agent recommendation system hitting 83% accuracy vs. **GPT-4** with a 1B model via **LoRA** fine-tuning (**Unsloth**), advised by Prof. Yongfeng Zhang

Partnerships Intern

Nas.io • New York, NY

Feb 2025 - May 2025

- Built and owned influencer outreach automation end-to-end, including an **Instagram scraper** and **Python/Klaviyo** pipeline that increased outreach 50x; also led client onboarding and partnerships, generating \$2K GMV in one month

PROJECTS

BYOA: Bring Your Own Agent

- Built a **Kubernetes** inference platform that deploys agents on arbitrary **Hugging Face** models behind OpenAI-compatible APIs, using **NVIDIA Dynamo** for disaggregated serving, KV-cache-aware routing, and GPU autoscaling

TaskCLI: SWE-bench-style evaluation runner

- Built a **Python** agent-evaluation harness for SWE-bench-style task bundles with reproducible **Docker** sandboxes, hidden-test patch grading, **JUnit XML** result parsing, **SQLite** artifact tracking, and OpenAI-compatible LLM/stub solvers

tochi.one: AI property management agents

- Built payment, lease-generation, and tenant-communication workflows using **LangGraph** and **MCP**, integrating **Stripe** and **Twilio** tools with persistent agent state

AdBlockIRL: AR ad-blocking proof-of-concept for Meta Quest

- Trained **YOLOv8** on **Roboflow** to detect 100+ concurrent ads at 5ms latency, applying real-time mosaic censoring via **OpenCV**; won \$2000 at Bain Capital Ventures AI Hackathon

EDUCATION

Rutgers University - New Brunswick

Bachelor of Science - Computer Science • Cum Laude

- Linux User Group (President), Road to Silicon Valley Program